TRM Encoder + SmolLM Cross-Attn Experiments (Overview)

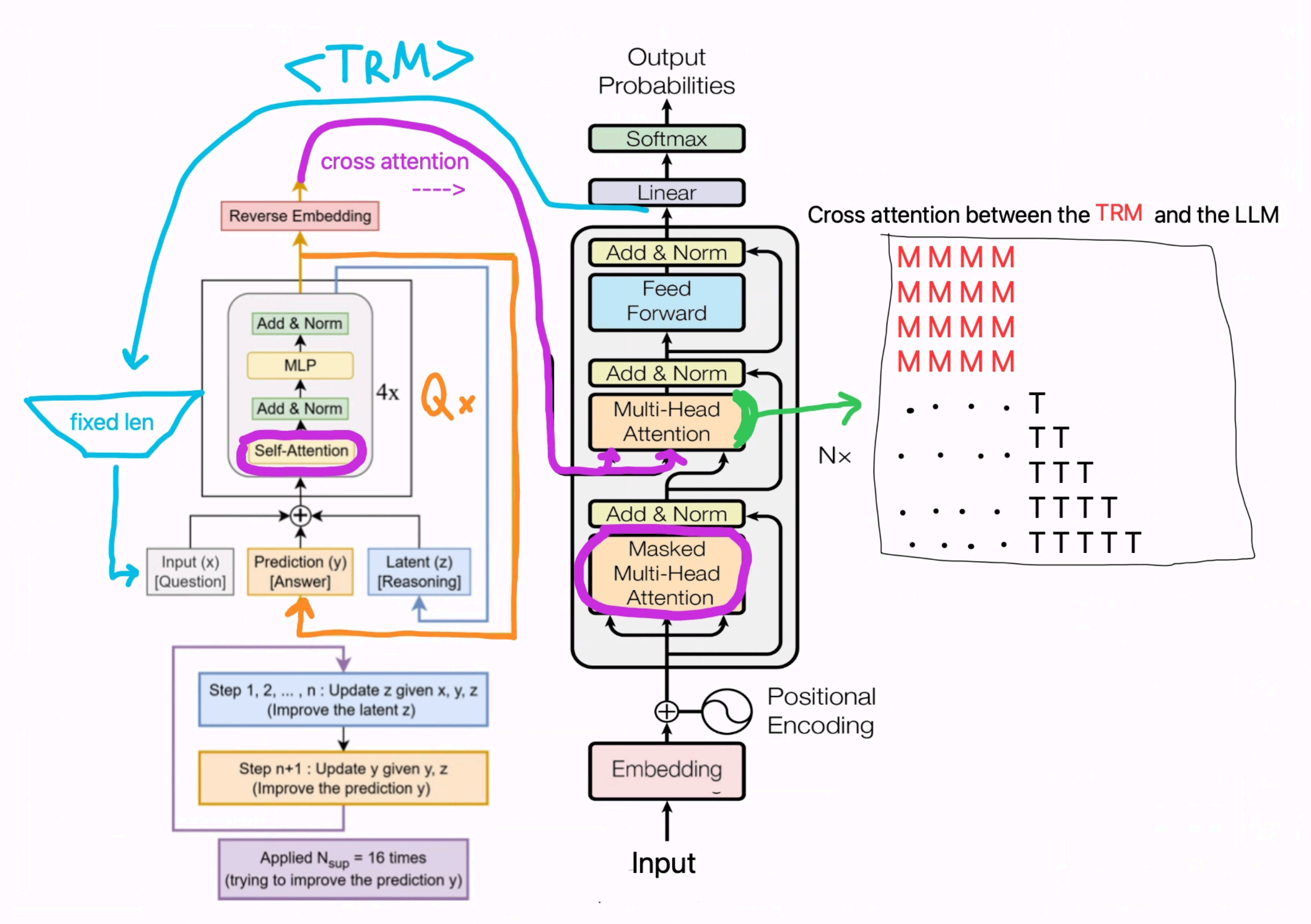

Figure 1: TRM encoder feeds a pretrained decoder via cross-attention.

The encoder compresses a long prompt into a fixed slot sequence, applies TRM recursive refinement, then expands to a decoder-length memory. The decoder receives this memory through cross-attention (optionally restricted to early layers) and we compare validation loss with cross-attention enabled vs disabled.

Experiment summary

Summary compares three setups: a short 30-layer baseline run, a longer run with larger gate init, and a recursive encoder run with cross-attn only in early layers and dropout.

Experiment	Key changes	Steps	Final cross	Final no-cross	Delta
Initial 30L	short run, encoder 512/128	200	1.5510	1.6283	+0.0773
Medium-long	gate init –2.0, encoder 512/128	2000	2.3770	2.3774	–0.0004
Recursive 5090	recursive Z, cross on layers 0-5, dropout 0.1	2000	2.0902	2.0902	+0.0000

Initial 30-layer run (200 steps)

Short 200-step run with 30-layer decoder, encoder 512/128, standard gates, no cross-attn dropout.

Table for the short-run setup described above.

Step	Cross	No-cross	Delta
50	1.8221	1.8156	–0.0065
100	1.9193	1.9388	+0.0195
150	1.4897	1.5367	+0.0470
200	1.5510	1.6283	+0.0773

Chart for the short-run setup described above.

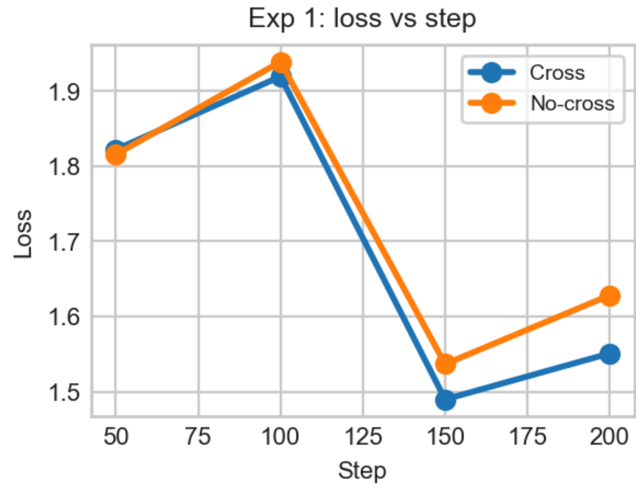

Figure 2: Exp 1: loss vs step.

Medium-long run (2000 steps)

Longer 2000-step run with 30-layer decoder, encoder 512/128, gate init –2.0, analytics on, no-cross eval.

Table for the medium-long setup described above.

Step	Cross	No-cross	Delta
200	2.2036	2.2415	+0.0379
400	2.4623	2.4785	+0.0162
600	1.7670	1.7717	+0.0047
800	1.8464	1.8467	+0.0003
1000	2.0721	2.0736	+0.0015
1200	2.1530	2.1517	–0.0013
1400	2.2641	2.2610	–0.0031
1600	2.2352	2.2352	+0.0000
1800	2.1117	2.1104	–0.0013
2000	2.3770	2.3774	+0.0004

Chart for the medium-long setup described above.

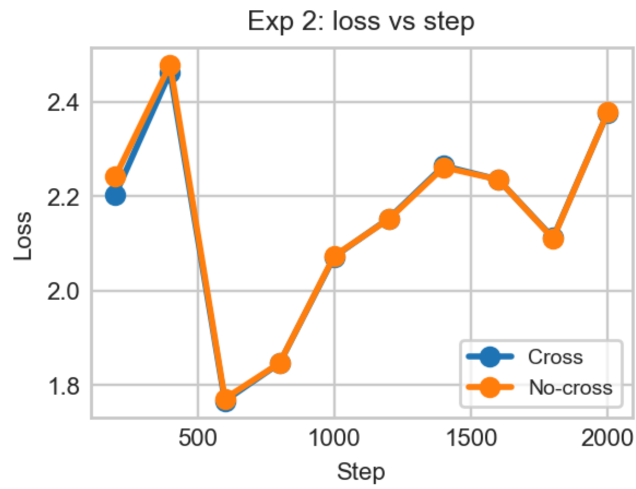

Figure 3: Exp 2: loss vs step.

Recursive 5090 run (2000 steps)

Recursive encoder updates (stride 64, short cycles 1/1, carry Z only), cross-attn limited to layers 0-5 with dropout 0.1.

Chart for the recursive setup described above.

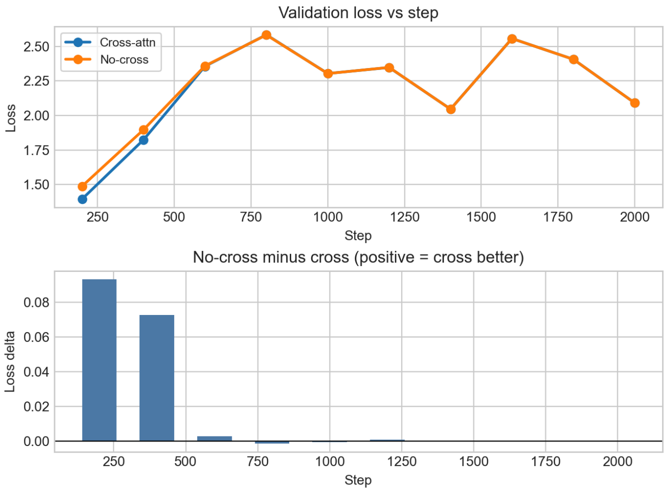

Figure 4: Recursive run validation loss with/without cross-attn.

Table for the recursive setup described above.

Step	Cross	No-cross	Delta
200	1.3943	1.4876	+0.0933
400	1.8237	1.8965	+0.0728
600	2.3558	2.3587	+0.0029
800	2.5846	2.5832	−0.0014
1000	2.3039	2.3034	−0.0005
1200	2.3475	2.3482	+0.0007
1400	2.0459	2.0458	−0.0001
1600	2.5572	2.5571	−0.0001
1800	2.4067	2.4066	−0.0001
2000	2.0902	2.0902	+0.0000

Interpretation

Across all runs, cross-attention provides a small early boost, then collapses to near-zero delta. The decoder learns to minimize loss while effectively ignoring cross-attn, even with recursive encoder updates and cross-attn restricted to early layers. This supports the hypothesis that a pretrained decoder will discard alien cross-attn to recover baseline next-token performance.